- $J_p, J_n = 0 \Rightarrow J_{p\,dri} + J_{p\,diff} = 0;\ J_{n\,dri} + J_{n\,diff} = 0$
- diff in p-side and n-side E_c / $E_v = qV_{bi}$
- $E_{fp} = E_{ip} - KT\ln(N_A/n_i)$ $\quad E_{Fn} = E_{in} - KT\ln(N_D/n_i)$

$$\Rightarrow E_{ip} - E_{in} = KT\ln\left(\frac{N_A N_D}{n_i^2}\right) = qV_{bi}$$

$$\Rightarrow V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = V_T \ln\left(\frac{p_{po}}{p_{no}}\right) = V_T \ln\left(\frac{n_{no}}{n_{po}}\right)$$

- $V_{bi}(GaAs) > V_{bi}(Si) > V_{bi}(Ge)$; V_{bi} can't be measured using voltmeter
- $p(x) = n_i \exp\left(\frac{E_i(x) - E_F}{KT}\right) = p_{po} \exp\left(\frac{E_i(x) - E_{ip}}{KT}\right)$
- Q_D → total uncompensated charge on each side

$$Q_D = AqN_A x_p = AqN_D x_n \qquad x_n/x_p = N_A/N_D$$

- $\mathcal{E}$ → Electric field $\quad \mathcal{E}_n(x) = \frac{qN_D}{\epsilon_{Si}}(x - x_n) \quad 0 < x < x_n$

$$V_{bi} = \frac{1}{2}|\mathcal{E}_{max}| \cdot W \qquad \mathcal{E}_p(x) = -\frac{qN_A}{\epsilon_{Si}}(x + x_p) \quad -x_p < x < 0$$

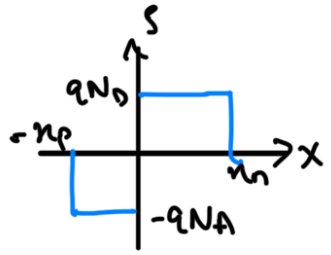
$W = x_p + x_n$ $\quad \mathcal{E}_{max} \rightarrow$ max $\mathcal{E}_p$ / $\mathcal{E}_n$ (@ $x = 0$)

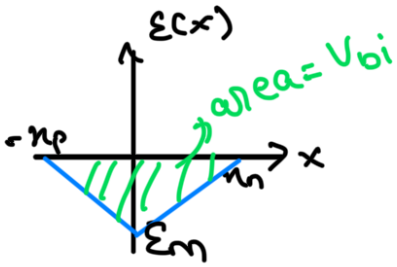
- V_n → potential dropped across n-side

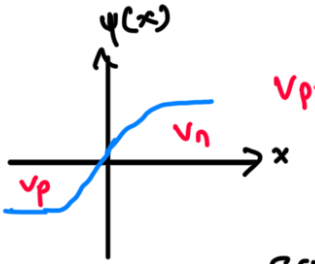
$$V_n/V_p = x_n/x_p = N_A/N_D \qquad x_p = \frac{N_D W}{N_A + N_D} \qquad x_n = \frac{N_A W}{N_A + N_D}$$

- $$W = \sqrt{\frac{2\epsilon_{Si} V_{bi}(1/N_A + 1/N_D)}{q}} = \sqrt{\frac{2\epsilon_{Si} V_{bi}}{qN_{eff}}} \qquad 1/N_{eff} = 1/N_A + 1/N_D$$

-

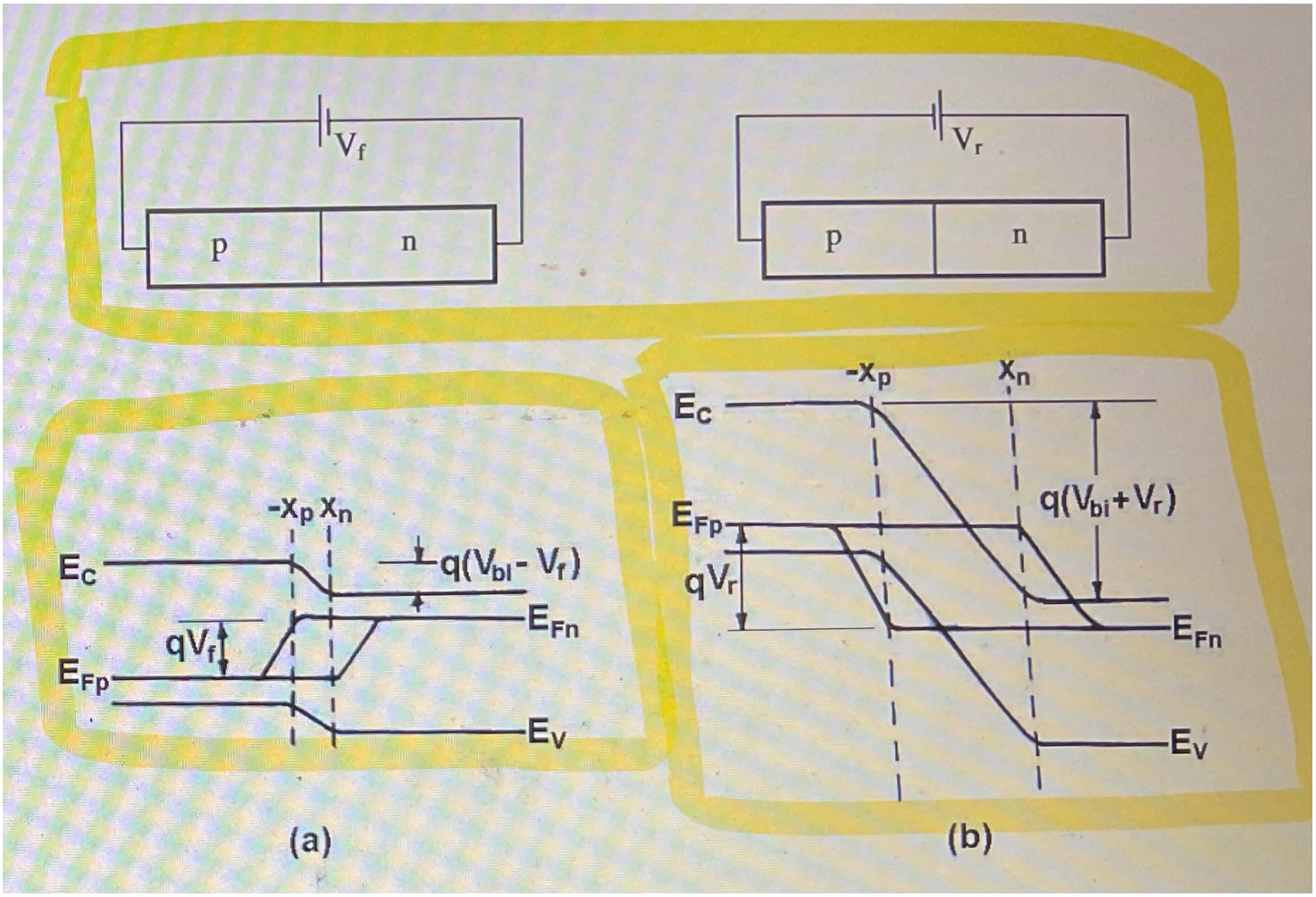






- linear graded function: $N_D - N_A = ax \Rightarrow \mathcal{E}(x) = \frac{qa}{2\epsilon_{Si}}\left(x^2 - \frac{W^2}{4}\right)$
- $\frac{d\mathcal{E}}{dx} = \frac{q}{\epsilon_{Si}}(N_D - N_A)$ → from poissons $\qquad V_{bi} = 2V_T \ln\left(\frac{aW}{2n_i}\right)$
- forward bias: $V = V_{bi} - V_f$ $\qquad$ reverse bias: $V = V_{bi} + V_r$

- $|E_m| = \frac{2(V_{bi}-V)}{W}$ $\quad W = \sqrt{\frac{2\epsilon_{si}(V_{bi}-V)}{qN_{eff}}}$



- C_j (junction capacitance) $\quad C_j = \frac{dQ_D}{dV} \quad Q_D = AqN_{eff}W$

$$C_j = A\varepsilon_{Si}\sqrt{\frac{qN_{eff}}{2\varepsilon_{Si}(V_{bi}-V)}} = \frac{\varepsilon_{Si}A}{W} = C_{j0}\left(1-\frac{V}{V_{bi}}\right)^{-1/2}$$

$p^+n \Rightarrow N_{eff} = N_D$

- $p_{ne} = p_{no}\exp(V/V_T) \qquad n_{pe} = n_{po}\exp(V/V_T)$

@ edge of depletion region

- long diode: $p_n - p_{no} = p_{no}\left[\exp(V/V_T)-1\right]\exp\left(-\frac{x-x_n}{L_p}\right)$

$$n_p - n_{po} = n_{po}\left[\exp(V/V_T)-1\right]\exp\left(\frac{x_1-x_p}{L_n}\right)$$

- $I = I_o\{\exp(V/V_T)-1\}$ $\quad I_o \rightarrow$ reverse saturation current

$$I_o = Aqn_i^2\left(\frac{D_p}{L_pN_D}+\frac{D_n}{L_nN_A}\right) \simeq \frac{qAD_np_{no}}{L_p}$$

- @ edge of depletion region: $n_p = n_i^2\exp(V/V_T)$
- $I_p(x=x_n) = Q_p/\tau_p \qquad I_p(x) = I_p(x_n)\exp\left(-\frac{x-x_n}{L_p}\right)$
- Ideality factor: $I \propto I_o\exp(V/\eta V_T)$ (real diode $V \gg V_T$)

$$I_{rec} = \frac{Aqn_iW}{2\tau}\exp\left(\frac{V}{2V_T}\right)$$ another component of current

- p^+n diode: $I(t) = \frac{Q_p(t)}{\tau_p} + \frac{\partial Q_p(t)}{\partial t} \qquad Q_p(t) = Aq\int_{x_n}^{\infty}(p_n-p_{no})\,dx$

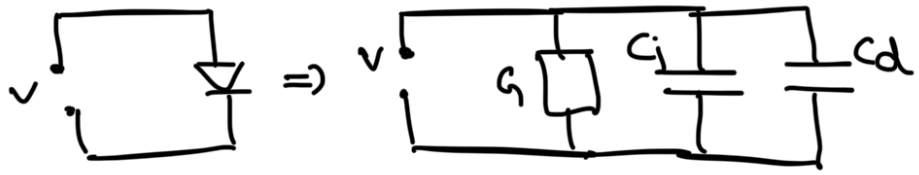
$$Q_p(t) = I\tau_p\exp(-t/\tau_p)$$

$$V(t) = V_T\ln\left[\frac{I\tau_p}{AqL_pp_{no}}\exp(-t/\tau_p)+1\right]$$

- t_{sd} (storage delay time) $= \tau_p\ln(1+I_F/I_R)$

- diffusion capacitance $C_d = \left(\frac{dI}{dV}\right)\tau_p = \frac{I}{V_T}\tau_p$
- τ_T (transit time) $= \frac{W_n^2}{2D_p} \Rightarrow C_d = G\tau_T$ (short base diode)

  $\tau_T = \tau_p$ (long base diode)



$G = I/V_T$

$C_j = A\sqrt{\frac{q\epsilon_{Si}N_D}{2(V_{bi}-V)}}$

$C_d = G\tau_T$

- Avalanche breakdown

  multiplication factor $M = \frac{n_{out}}{n_{in}} = \frac{1}{1-P} = \frac{1}{\left[1-\left(\frac{V_R}{V_{BR}}\right)^n\right]}$

  $V_{br} = \frac{\mathcal{E}_m W}{2}$

- $I_p(x_n) = \frac{AqD_p p_{no}}{L_p}\left[\exp(V/V_T) - 1\right]$

$I_n(-x_p) = \frac{AqD_n n_{po}}{L_n}\left[\exp(V/V_T) - 1\right]$

Short base diode $\Rightarrow$ replace $L \rightarrow W$